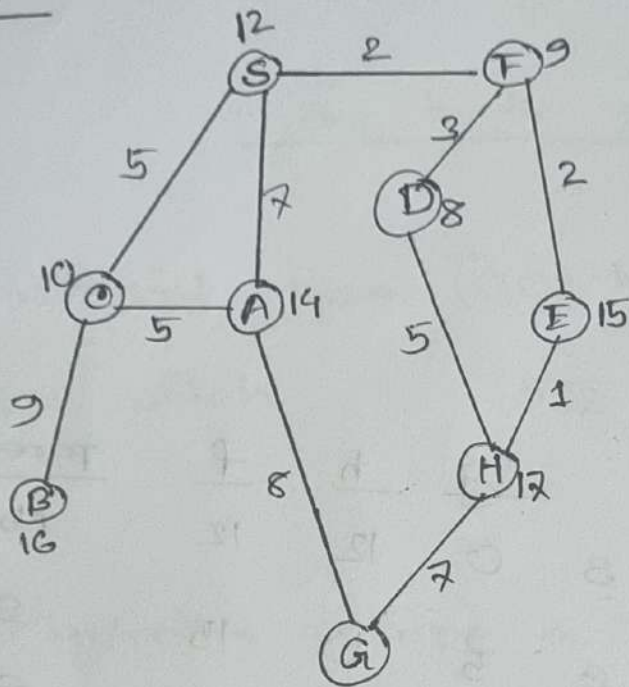


Ans. to the que. no. 1

QBFS



Visited: S F D C A

PQ: ~~S~~ ~~F~~ ~~A~~ ~~D~~ ~~E~~

	<u>h(n)</u>	<u>parent</u>
S	0	None
F	9	S
A	14	S
C	10	S
D	8	F
E	15	F
H	17	D
B	16	C
G	0	A

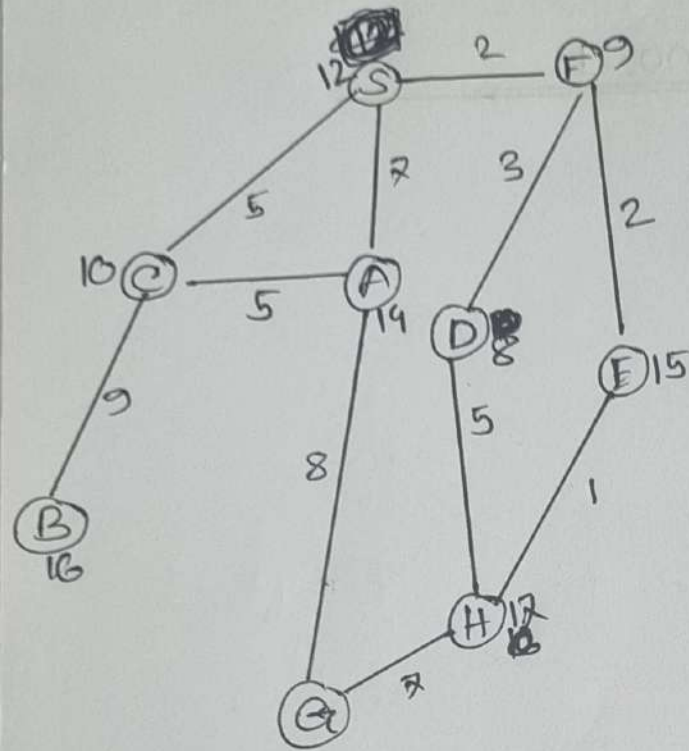
So, path

$S \rightarrow A \rightarrow G$

Total cost $(7+8) = 15$

Ans.

A-star Search



visited: S F D C E A

PQ: ~~S~~ ~~C~~ ~~A~~ ~~D~~ ~~E~~ H B

path → S → A → G

cost → 15

	<u>g</u>	<u>h</u>	<u>f</u>	<u>parent</u>
	0	12	12	None
S	0	12	12	None
C	5	10	15	S
A	7	14	21	S
F	2	9	11	S
D	5	8	13	F
E	4	15	19	F
H	10	12	22	E
B	14	16	30	C
G	15	0	15	A

Both are same, but GBFS only checked the heuristic, which will give bad result most of the time. Also, in this graph there are many nodes which have bad h, and the graph is not

admissible. That's the reason we didn't get the optimal path ~~in~~ in both cases.

Ans. to the que. no. 2

Actual cost from $(0,0)$ to $(5,5)$ is 5 diagonal boxes, which is $5\sqrt{2} \approx 7.07$

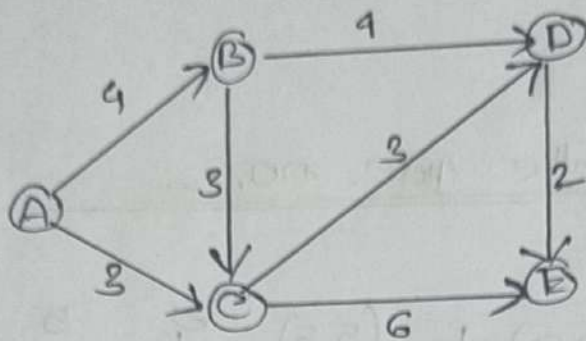
$$\text{Manhattan distance, } h_1 = |5-0| + |5-0| \\ = 10$$

$$\text{Euclidean distance, } h_2 = \sqrt{5^2 + 5^2} \\ = 5\sqrt{2} \approx 7.07$$

$$h_2 \leq \text{Actual cost.}$$

Euclidean distance, h_2 will expand fewer nodes in this grid.

Ans. to the que. no. 3



Adm to design a admissible heuristic table for
this graph we need to consider,

$$h \leq \text{Actual cost}$$

so, Actual ~~cost from~~ shortest cost,

$$A \rightarrow C \rightarrow D \rightarrow E = 8 \text{ (A)}$$

$$B \rightarrow D \rightarrow E = 6 \text{ (B)}$$

$$C \rightarrow D \rightarrow E = 5 \text{ (C)}$$

$$D \rightarrow E = 2 \text{ (D)}$$

$$E = 0 \text{ (E)}$$

Node	heuristic (h)
A	8
B	6
C	5
D	2
E	0

Ans. to the que. no. 4

From every corner,

Top left, 1 (neigh - 4, 6, 10)

max 10, so now on 10 (neigh - 4, 6, 2, 9, 8, 5, 3)

No higher value, stuck at local maxima

Bottom left, 4 (neigh - 5, 6, 7)

max 7, so now (N - 5, 3, 1, 8, 6)

max 8, now (N - 1, 9, 8, 4, 15, 3, 6)

max 15 reached global maxima

Bottom right, 10 (N - 5, 3, 6)

stuck at local maxima

Top right, 2 (N - 5, 3, 1)

max 5, now (N - 3, 4, 10, 9, 8, 4, 1)

max 10, stuck at local maxima

~~Ans~~ For this grid, only if I start from bottom

left I can achieve highest value. That's why

Hill climbing is not a good option when

there are many local maxima or minima

as it's eventually stuck in one of the local maxima/minima. (Ex. other corners)
For this reason, ~~it is~~ all corners can't get highest value.

Ans. to the Que. no. 5

If, $[\Delta E = f(N) - f(e)]$

$\Delta E > 0$; choose Neighbours

$\Delta E < 0$; Accept with probability

Now, $e^{-\frac{\Delta E}{T}}$; where when T is high, means probability is high; accepting worse moves,

but when T (temperature) is low, means low possibility of accepting worse moves.

That's how temperature (T) parameter controls the probability of neighbours selection in simulated annealing.

Ans. to the que. no. 2

Encoding Scheme No duplicate cities, first and last node will be same.

Ex:- A B C D E F G H A

Parent: Potential 6 parents,

A C D E F G B H A (49)

B C G H A F E D B (39)

C A G H B F D E C (38)

A G H B F E D C A (39)

B G H F D C E A B (48)

G H F C D E A B G (49)

Fitness: $\frac{1}{\text{total dist.}}$ the (shorter distance, higher fitness) [no edge = α]

Crossover and Mutation:

$P_1 \rightarrow A \ C \ \boxed{D \ E \ F \ G} \ B \ H \ A$

$P_2 \rightarrow B \ C \ G \ H \ \boxed{A \ F \ E \ D} \ B$

$C_1 \rightarrow B \ C \ \boxed{D \ E \ F \ G} \ H \ A \ B$

$C_2 \rightarrow C \ G \ B \ H \ \boxed{A \ F \ E \ D} \ C$

Now mutation, (Randomly swap two cities)

$C_1 \rightarrow B \ C \ D \ H \ F \ G \ E \ A \ B$

$C_2 \rightarrow C \ G \ B \ H \ A \ E \ F \ D \ C$

$$C_1 \text{ cost} \rightarrow (2+9+7+8+2+3+7+4) = 42 \Rightarrow \frac{1}{42}$$

$$C_2 \text{ cost} \rightarrow (6+8+5+9+7+6+4+9) = 54 \Rightarrow \frac{1}{54}$$

After one iteration C_1 is the best solution among the childs,

Next Population, (2-elitism)

~~$P_3 \rightarrow C A B H A F E D$~~

$P_3 \rightarrow C A G H B F D E C$

$P_2 \rightarrow C G B H A F E D B$

$C_1 \rightarrow B C D H F G E A B$

$C_2 \rightarrow C G B H A E F D C$

Ans. to the que. no. 8

Given func,

$$x^4 - 14x^2 + 24x + 30$$

Encoding Scheme:- ~~To do~~ We need to do

crossover and mutation, so binary encoding

would be helpful. So, let x is represented as a binary string ($-15 \leq x \leq 15$)

Population: [first bit + (0) - (1)]

$$P_1 \quad 00101 \rightarrow 5 \quad f(x) = 425$$

$$P_2 \quad 10111 \rightarrow -2 \quad f(x) = 1522$$

$$P_3 \quad 00111 \rightarrow 2 \quad f(x) = 1913$$

$$P_4 \quad 11111 \rightarrow -15 \quad f(x) = 42145$$

Fitness: Shift binary to decimal, first bit is sign bit [0 $\rightarrow$ (+), - $\rightarrow$ (1)]. Now get

the decimal value and just put into the

function and then $\frac{1}{1000000 + f(x)}$ evaluate.

Max is better on global minimum.

Crossover

(P_1, P_2)

$P_1 \rightarrow 001101$
 $P_2 \rightarrow 10111$

$C_1 \rightarrow 00111$

$C_2 \rightarrow 10101$

(P_3, P_4)

$P_3 \rightarrow 001111$

$P_4 \rightarrow 11111$

$C_3 \rightarrow 00111$

$C_4 \rightarrow 11111$

Now, ~~if~~ Mutation, flip one random bit,

$C_1 \rightarrow 01111 \rightarrow 15 \Rightarrow f(x) = 47865$

$C_2 \rightarrow 10001 \rightarrow 1 \Rightarrow f(x) = 41$

$C_3 \rightarrow 00101 \rightarrow 5 \Rightarrow f(x) = 425$

$C_4 \rightarrow 01111 \rightarrow 15 \Rightarrow f(x) = 47865$

best solution, after iteration, C_1 and C_4 both. $\left(\frac{1}{100000 + 47865} \right)$

Next population,

C_1, C_4 and then C_3, C_2

Ans. to the que. no. 9

Encoding and Parent: There will be no duplicated or repetition guests/nodes. Now, the

4 parents' potential parents are,

$$P_1 \rightarrow G1, G2, G3, G4, G5, G6, G7, G8$$

$$P_2 \rightarrow G2, G1, G4, G5, G7, G8, G6, G3$$

$$P_3 \rightarrow G8, G7, G6, G5, G4, G1, G3, G2$$

$$P_4 \rightarrow G6, G7, G8, G5, G3, G2, G1, G4$$

Fitness: So, for preferred neighbor +1 and for -1 for avoided neighbour. Also, table is circular so all guests preference will count.

$$P_1 \rightarrow (1+1+1+2+1+1+1) = 8$$

$$P_2 \rightarrow (1+1-1+1+1+1+1+1) = 7$$

$$P_3 \rightarrow (1+1+1+2+1+1-1+1) = 7$$

$$P_4 \rightarrow (1+1+1+1-1) = 3$$

two fittest chromosomes are P_1, P_2

No, regular crossover will not work in this scenario. Cause, there have to be no duplicate guests meaning all have to be unique and everyone should appear once.

Just, randomly swap between two random indexes.

Example,

$C_1 \rightarrow G_4 G_1 G_2 G_3 G_5 G_7 G_6 G_8$

Mutated $C_1 \rightarrow G_4 G_1 \underline{G_3} \underline{G_2} G_5 G_7 G_6 G_8$

Ans. to the que. no. 10

① True, As time passing, the T is keep increasing decreasing. Means, when it's denying worse solution as time passes or accepting only good solution.

② False SA is ~~more~~ not always guaranteed better in all cases (depends on landscape). Also, in some simple or smooth search spaces hill climbing performs even better.

③ False High temperature means random search. Actually ~~low~~ when temperature is very low the SA $\approx$ Hill climbing.

④ False Random restarts helps escape local maxima, increase the chance of finding global optimum.

⑤ False Larger iterations increase, not eliminate the chance to find the global optimum (though not guaranteed in all cases).

Ans. to the que. no. 11

a) Encoding: 20 distinct areas, 3 cameras, so camera's exist in will be place as 1 and empty space areas 0. But there can be only 3 camera's (1) ~~the~~ last camera will be always the powerful one.

b) Fitness: Each unique area's covers (+1). Max number, better the answers. So, the better the placement, the more areas covered, the higher the fitness.

c) Parents,

001000001000000100000000 → 12

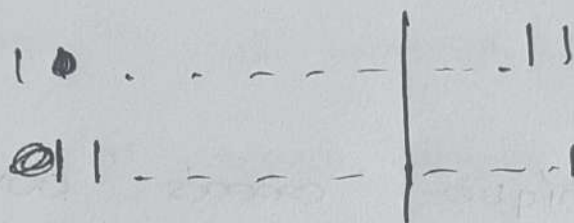
111000000000000000000000 → 6

00001000000000100000000001 → 14

000000000000001000000001001000 → 11

④ No, single point crossover will not work

If we do single point crossover there would be a chance for 4 cameras on more in a single parent which is simply not possible. Example,



Now, one will have 4 cameras and other have only 2. So single point crossover won't work for this example.

Ans. to the que. no. 12

a) Using a binary string where 0 means group 1 and 1 means group 2. And serial will

$A_1, A_2, \dots, A_6$. Ex: 100111, 111000

$$\left[\begin{array}{cccccc} 1 & 1 & 1 & 0 & 0 & 0 \\ A_1 & A_2 & A_3 & A_4 & A_5 & A_6 \end{array} \right] \rightarrow (A_1, A_2, A_3) (A_4, A_5, A_6)$$

b) Just need to get the value of A_1 to A_6 and different groups are maximized and score between same group is minimized.

$$\text{fitness} = \text{cross sum} + \frac{1}{\text{same sum}}$$

$$111000 \rightarrow (A_1, A_2, A_3) (A_4, A_5, A_6)$$

$$\begin{aligned} \text{cross sum} &= (6+5+2) + (4+2+3) + (9+6+2) \\ &= 44 \end{aligned}$$

$$\begin{aligned} \text{same sum} &= (8+3+2) + (2+1+5) \\ &= 31 \end{aligned}$$

$$\text{So, fitness} \rightarrow \frac{1}{31} + 44 = 44.03$$

So, max the number better the parent.

③ We can use single point crossover

for this example,

Suppose,

$$P_1 \rightarrow 001101$$

$$P_2 \rightarrow 111000$$

$$C_1 \rightarrow 001000$$

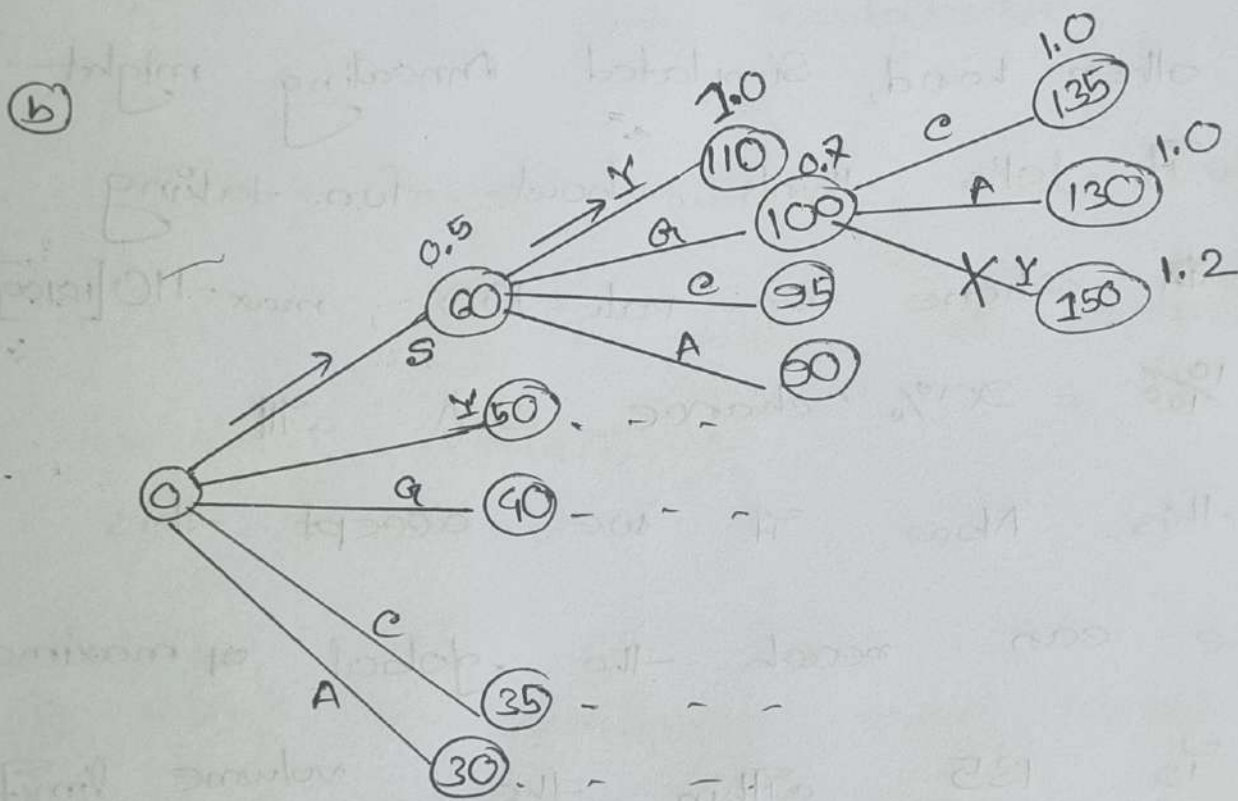
$$C_2 \rightarrow 111101$$

$$\text{So, } G_1 \rightarrow (A_1 A_2 A_4 A_5 A_6) (A_3)$$

$$G_2 \rightarrow (A_1 A_2 A_3 A_4 A_6) (A_5)$$

Ans. to the que. no. 13

① Binary encoding. Starting with empty lunchbox $[0,0,0,0,0]$ $\underline{S}, \underline{A}, \underline{Y}, \underline{Q}, \underline{C}$. then adding 1 item one by one and checking the fitness. Ex: $[10000]$ then $[11000], [10100]$ etc. while staying in the 1 liter limit.



$[10100]$ (hill climbing)

Neighbouring solutions for hill climbing method where max number is 110, but actually the most/optimal max number is 135.

© If we use S.A. instead of H.C. then

$[10010]$ also ~~can~~ been ~~ex~~ accepted, because,

hill climbing rejects this neighbour immediately because it's worse (lower nutrition value)

On the other hand, Simulated Annealing might accept it. Let's think about for taking $[10010]$ where the $val = 100$, $max = 110$ $[10100]$

So, $e^{-\frac{10}{100}} \approx 90\%$ chance S.A. will accept this. Now if we accept this then we can reach the global ~~at~~ maxima which is 135 within the volume limit.

Ans. to the que. no. 14

h_1, h_2 are admissible and h_3 is not admissible,

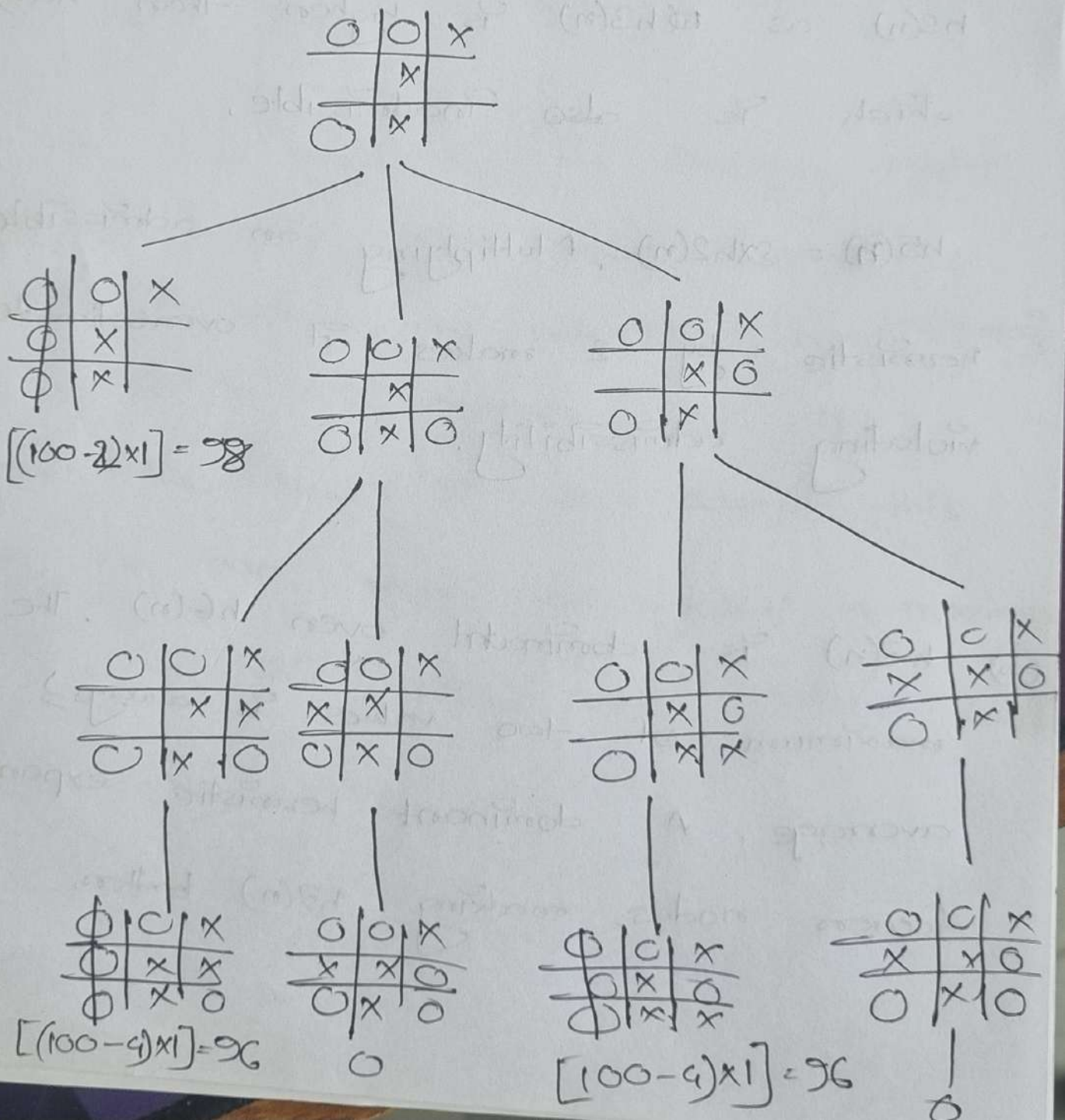
(a) $h_9(n) = \max(h_2(n), h_3(n))$, Max will be $h_3(n)$ as $h_3(n)$ is higher than $h_2(n)$ which is also inadmissible,

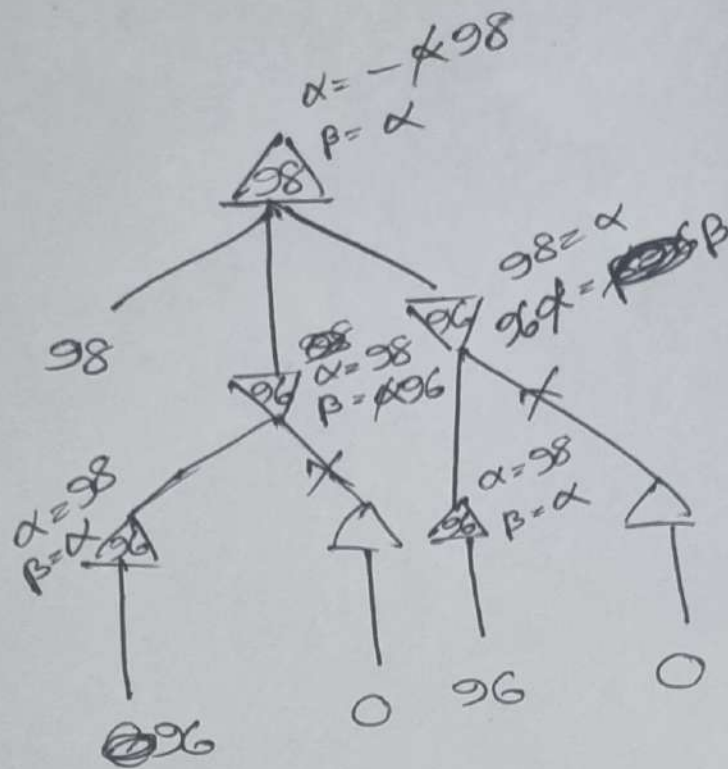
$h_5(n) = 2 \times h_2(n)$, Multiplying an admissible heuristic by 2 makes it overestimate, violating admissibility.

(b) $h_7(n)$ is dominant over $h_6(n)$. The maximum of two values is always $\geq$ their average. A dominant heuristic expands fewer nodes, making $h_7(n)$ better.

© $h_2(n)$ is the best choice. As $h_2(n)$ will be the most dominant heuristic

Ans. to the que. no. 15





$\Rightarrow 98$